Syllabus for Ph. D. in Computer Science
(PhD-032) Advance Research Studies in Computer Engineering

Module-I (Data Warehouse and Mining)

Introduction to Data Warehousing. Client/Server Computing mode, Parallel processors & Cluster Systems. Data Warehousing Components, Building a Data Warehouse, Mapping the Data Warehousing to a Multiprocessor Architecture, DBMS Schemas for Decision Support. Introduction to Data Mining. Decision Trees, Neural Networks, Nearest Neighbor & Clustering, Genetic Algorithms, Rule Induction, Selecting & Using the Right Technique.

Module-II (Advance Computer Network and Ad-hoc Networks)

Next Generation IP protocol, TCP extensions for high speed network, Introduction to SCTP. P2P file sharing and structure overlay network. Introduction to wireless and ad-hoc networks, Mobility in networks, Mobile IP, Mobile TCP, advantages and limitations of ad-hoc networks, Routing in MANETs.

Module-III

Introduction to MatLab and NS-2.

1. SOFTWARE PROJECT MANAGEMENT (PhDCS 103)

MODULE I

Overview of Project Management, PMI Processes, Software project phases, Organizational structures, Project charter, Statement of Work (SOW) **Planning Phase,** Development lifecycle models, Matching lifecycles to projects, Project plans, Work Breakdown Structures (WBS) **Estimation and Budgeting**, Estimation, Budgeting, Project, selection, NPV, ROI, Payback models

MODULE II

Scheduling, Project network diagram fundamentals, PERT techniques, Gantt charts, Critical chain scheduling **Risk and Change Management**, Mid-term review , Risk management, Change control, More MS-Project.

Development Management, Team models, Requirements process, Configuration, management, Software metrics, Programming languages & tools, Managing conflict and motivating, MS-Project: Assigning Resources.

MODULE III

Project Control, Status reporting, Project metrics, Earned value analysis, Communications Techniques, Process Improvement, MS Project: (a) Resource leveling (b) Other views

System Test Process, Test specifications, Black box and white box testing, Test scripts, Unit and integration testing, Acceptance test specifications, Test tools, MS Project:(a) Reporting.

Final Phases & Other Issues, Project Recovery, Documentation, Cutover/Migration Post Project Reviews, Closing, MS Project: (a) Advanced features.

Reference Books:

1. S. McConnell, "Software Project Survival Guide" (1997)

2. S. Berkun, "The Art of Project Management", (2005)

3. C. Larman, "Agile and Iterative Development: A Manager's Guide", (2003)

4. W. Royce, "Software Project Management: A Unified Framework", (1998)

5. J. Highsmith, "Agile Project Management: Creating Innovative Products", (2004)

6. T. DeMarco, "The Deadline: A Novel About Project Management", (1997)

7. T. DeMarco, "Peopleware: Productive Projects and Teams", (1999)

8. E. Bennatan, "On Time Within Budget: Software Project Management Practices and Techniques", (2000)